MILITARY SPECIFICATION

CHARGE, PROPELLING, 175MM, M199

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Propelling, M199, for 175MM Gun, M113 and M113A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	- Powder, Black
MIL-A-2550	- Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1320

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DRAWINGS

ARMY

7548187	- Packing and Marking of Metal Containers for Propelling Charges
8837922	- Igniter Core Loading Assembly
8837923	- Igniter Core Assembly
9246587	- Base Igniter Charge Loading Assembly
8868757	- Gage, Chamber
9269082	- Packing for Container, Metal, Ammunition, M460A2 for Charge, Propelling, 175MM, M199, for Gun, M113
9266755	- Charge, Propelling, 175MM, M199, for Gun, M113

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed y the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Propelling Charge Assembly.-The propelling charge assembly shall comply with all requirements specified on Drawing (dwg.) 9266755, and with all requirements specified in applicable specifications.

3.3 Moisture Content of Black Powder.-The moisture content of black powder at the loading station, at the time of loading, shall not exceed the amount specified in Specification MIL-P-223.

3.4 Propelling Charge Increment Weight.-The propellant weight shall comply with the weight specified in the loading authorization within the following tolerances: Increments Number (No.) 1 Charge, plus or minus (±) 0.40 ounces; Increment No. 2 Charge, ± 0.30 ounces.

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3.5 Profile and Alignment (See 4.2.2.5).-The completed propelling charges shall freely enter a profile and alignment gage, which will be referred to herein as a chamber gage.

3.6 Proving Ground.-The muzzle velocity, pressure and standard deviation of the velocity shall be as specified in Table I, when tested as specified in 4.3.3.

TABLE I

Temperature (°F)	Velocity* (fps) Min.	Max.	Standard Deviation	Maximum Pressure (psi)
+70	2285	2335	10	58000

*Velocity for information only.

(fps) - feet per second
(°F) - degrees Fahrenheit
(min.)- minimum
(max.)- maximum
(psi) - pounds per square inch

3.7 Assembly Torque (See 4.2.2.7).-The test plug shall be assembled to withstand subsequent application of the min. assembly torque specified on the applicable drawings, without moving.

3.8 Workmanship.-All components shall be free of dirt, grease and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government Verification.-Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify, at unscheduled intervals, all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "Critical" be accepted solely on the basis of the contractor's records.

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4.2 Quality Conformance Inspection

4.2.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or subassemblies other than the items of delivery, shall be homogeneous and of a size convenient to the contractor. Each lot shall contain:

a. Propellant from not more than one lot.

b. Black powder of one interfix lot number from one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent (%). Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government.

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Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40% for each Major Defect and an AQL of 0.65% for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100% inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Igniter Core Loading Assembly (See dwg. 8837922).

Categories	Defects	Method of Inspection
Critical: None defined.		
Major:	AQL 0.40%	
101.	Assembly damaged to the extent that black powder can escape.....	Visual
Minor:	AQL 2.5%	
201.	Cloth improper color.....	Visual
202.	Any seam or opening incompletely stitched.....	Visual
203.	Threads broken.....	Visual
204.	Tabs missing or incomplete.....	Visul
205.	Evidence of poor workmanship (See 3.8).....	Visual

4.2.2.2 Igniter Core Assembly (See dwg. 8837923).

Categories	Defects	Method of Inspection
Critical:		
1.	Igniter core loading assembly missing.....	Visual
Major: None defined.		
Minor:	AQL 1.5%	
201.	Igniter core loading assembly not properly tied (when applicable).....	Visual
202.	Igniter core loading assembly not properly located.....	Visual
203.	Evidence of poor workmanship (See 3.8).....	Visual

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4.2.2.3 Increment No. 1 Loading Assembly (See dwg. 9276581 , covering a detail of dwg. 9266755).

Categories	Defects	Method Of Inspection
Critical:		
1.	Igniter core or igniter charge assembly missing	Visual
Major:	AQL 0.40%	
101.	Assembly damaged to the extent that black powder or propellant can escape	Visual
Minor:	AQL 2.5%	
201.	Any seam or opening incompletely stitched	Visual
202.	Threads broken	Visual
203.	Lacing not tied at both ends	Visual
204.	Igniter core assembly not centered	Visual
205.	Marking misleading or unidentifiable	Visual
206.	Evidence of poor workmanship (See 3.8)	Visual
207.	Ties of igniter core not securely sewed to body	Visual

4.2.2.4 Increment No. 2 Loading Assembly (See dwg. 9276582 , covering a detail of dwg. 9266755).

Categories	Defects	Method Of Inspection
Critical: None defined.		
Major:	AQL 0.40%	
101.	Igniter core assembly missing	Visual
102.	Assembly damaged to the extent that propellant can escape	Visual
Minor:		
201.	Any seam or opening incompletely stitched	Visual
202.	Threads broken	Visual
203.	Lacing not tied at both ends	Visual
204.	Igniter core assembly not centered	Visual
205.	Marking misleading or unidentifiable	Visual
206.	Evidence of poor workmanship (See 3.8)	Visual

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4.2.2.5 Propelling Charge Assembly (See dwg. 9266755).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:	AQL 0.65%	
101.	Chamber gage failure	Gage 8868757
102.	Total length, max	Gage CD
103.	Assembly damaged to the extent that black powder or propellant can escape	Visual
Minor:	AQL 1.5%	
201.	Increments insecurely tied at junctures	Visual-Manual
202.	Ties loose or broken	Visual-Manual
203.	Marking misleading or unidentifiable	Visual
204.	Evidence of poor workmanship (See 3.8)	Visual

4.2.2.6 Metal Container Prior to Tightening Screw Assembly (See dwg. 9269082).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:	100% Inspection	
101.	Gasket missing, improperly assembled or damaged to the extent that it will not pass air pressure test	Visual
Minor:	AQL 1.5%	
201.	Any packing component missing	Visual
202.	Cover cannot be removed by hand	Visual-Manual
203.	Charge cannot be removed by hand	Visual-Manual

4.2.2.7 Metal Container, Sealed (See dwg. 9269082 and 7548187).

Categories	Defects	Method Of Inspection
Critical:	None defined.	

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Major: 100% Inspection
101. Test plug missing.....Visual

Major: AQL 0.25%
102. DOD symbol misleading or unidentifiable..Visual
103. Test plug fails to withstand min. assembly torque (See 3.7).....Test

Minor: AQL 2.50%
201. Test plug not greased.....Visual
202. Car seal missing or incorrectly applied..Visual
203. Cover incompletely engaged.....Visual
204. Marking misleading or unidentifiable.....Visual

4.2.2.8 Metal Container with overpack (see dwg. 9219182 covering a detail of dwg. 9269082).

Categories Defects Method Of Inspection

Critical: None defined.

Major: None defined.

Minor AQL percent
201. Nails missing.....Visual
202. Strapping missing or loose.....Visual
203. Marking missing, misleading or unidentifiable.....Visual

4.2.3 Testing

4.2.3.1 Moisture Content of Black Powder (See 3.3) - Major Defects.-The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification, at least one (1) sample of black powder from each eight (8) hours production of igniters. If the sample fails to comply with the applicable requirement, the eight (8) hours production represented by the sample shall be rejected. The test shall be performed as specified in 4.3.1.

4.2.3.2 Propelling Charge Increment Weight (See 3.4); Weight of Black Powder for Igniter Core Assemblies (See dwg. 8837922); and Weight of Black Powder for Igniter Base Charges (See dwg. 9246587 - Major Defects.-The charges shall be weighed and check-weighed 100%. The check weighing shall be accomplished independently of the original weighing, using a different balance from that used to make the original weighing and, if performed manually, shall be performed by another operator. Any charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. CD

ITEM

Increment 1
Increment 2
Igniter Core
Igniter Base

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4.2.3.3 Air Pressure Test (See dwg. 7548187) - Major Defect.-The container shall be checked 100% as specified in 4.3.2. If any container fails to meet the requirements, it shall be rejected and removed from the lot. CD

4.2.3.4 Proving Ground (See 3.6) - Muzzle Velocity and Muzzle Velocity Standard Deviation - Major Defects.-The sampling plan for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B3, Code Letter D and an AQL of 0.65%. In addition, if the standard deviation of the sample multiplied by the factor 0.65 exceeds the applicable lot requirement, the lot shall be rejected.

4.2.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 and 4.2.3 (See 6.2).

4.2.4.1 Government Use of Contractor's Inspection Equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test Methods and Procedures

4.3.1 Moisture

4.3.1.1 Black Powder.-The moisture content of the black powder shall be determined in accordance with MIL-P-223.

4.3.2 Air Pressure Test for Container.-The container shall be subjected to an internal pressure of 3 psi. A pressure drop equivalent to 3 inches of water in 15 seconds shall be cause for rejection.

4.3.3 Proving Ground.-These tests shall be conducted at a Government proving ground in accordance with Acceptance Test Procedures (See 6.3). Five (5) charges shall be conditioned at +70°F for a minimum of 24 hours and fired at zone 2, using inert-loaded projectiles and dummy fuzes.

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5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A.-The propellant charges shall be packed in accordance with dwg. 9269082. Containers shall be closed, sealed, tested and marked in accordance with the requirements of dwg. 7548187.

5.2 Marking.-Marking shall be in accordance with dwg. 7548187.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

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6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (See 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion data for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 Submission of Test Data.-In addition to normal distribution of records, when the charges are procured by the Department of the Army, one copy of all ballistic test data will be forwarded to each of the following: Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-QA-A-A and SMUPA-AD-E-P.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1320-A-141

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